How to reach 94% Accuracy on CIFAR-10

A Deep Learning Approach

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CIFAR-10 Dataset

- 60,000 32x32 color images in 10 classes
- 50,000 training images and 10,000 test images
- Current state-of-the-art accuracy: 99.5%[1]
- My goal: Achieve at least 90% accuracy using an AlexNet inspired CNN

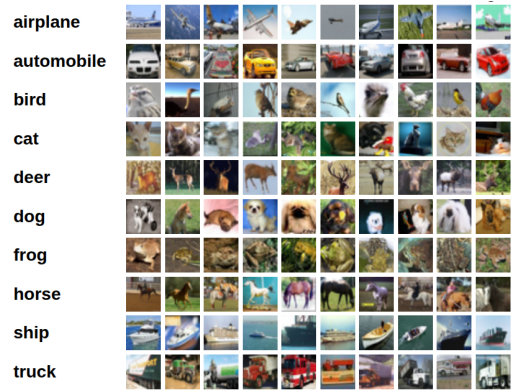


Figure: Sample images from CIFAR-10 dataset[2]

[1] Dosovitskiy et al. 2021.
[2] Krizhevsky 2009.

Model Architecture

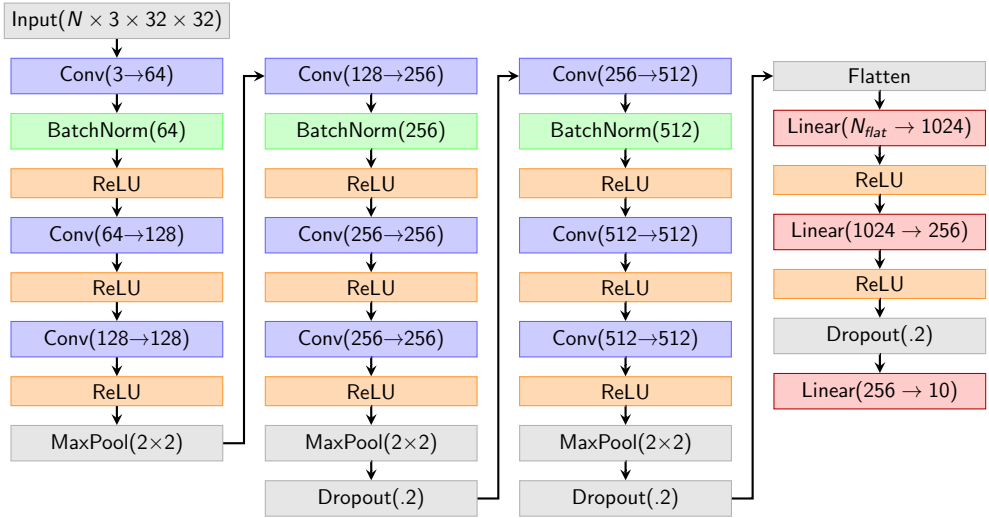


Figure: CNN architecture inspired by AlexNet

Training setup

- **Framework:** PyTorch
- **Optimizer:** SGD with learning rate 0.01, momentum 0.9, weight decay $5 \cdot 10^{-4}$
- **Loss function:** Cross-Entropy Loss
- **Batch size:** 128
- **Epochs:** 100
- **Hardware:** NVIDIA GTX 3060, 12GB VRAM
- Logging with MLFlow[3]
- Model analysis with FiftyOne[4]

$\Rightarrow$ 88.3% accuracy after 100 epochs



Figure: NVIDIA GTX 3060[5] used for training

[3] https://mlflow.org/
[4] https://voxel51.com/fiftyone/
[5] https://www.nvidia.com/de-de/geforce/graphics-cards/30-series/rtx-3060-3060ti/

Data Augmentation

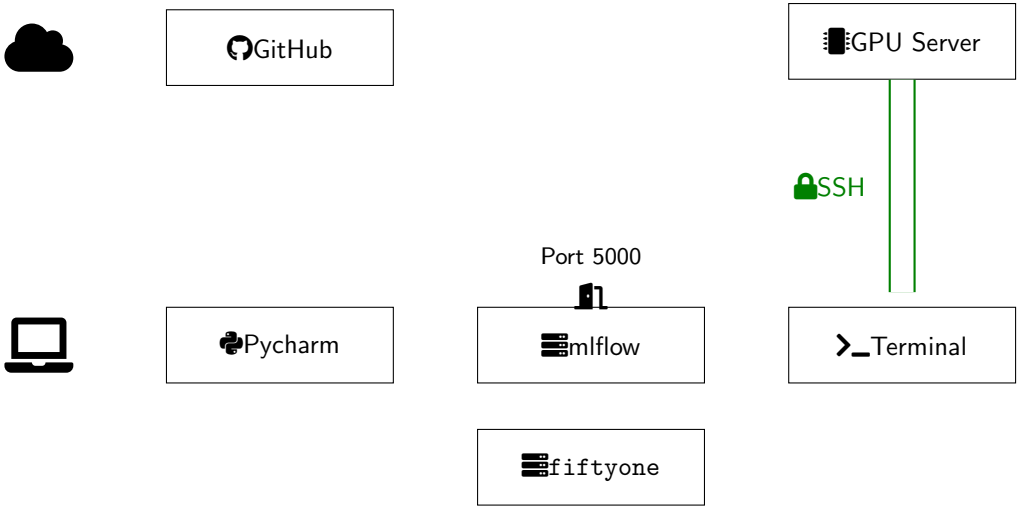
- Color Jitter (brightness, contrast, saturation)
- Perspective Transform
- Random Horizontal Flip

$\Rightarrow$ 92.5% accuracy after 100 epochs with data augmentation

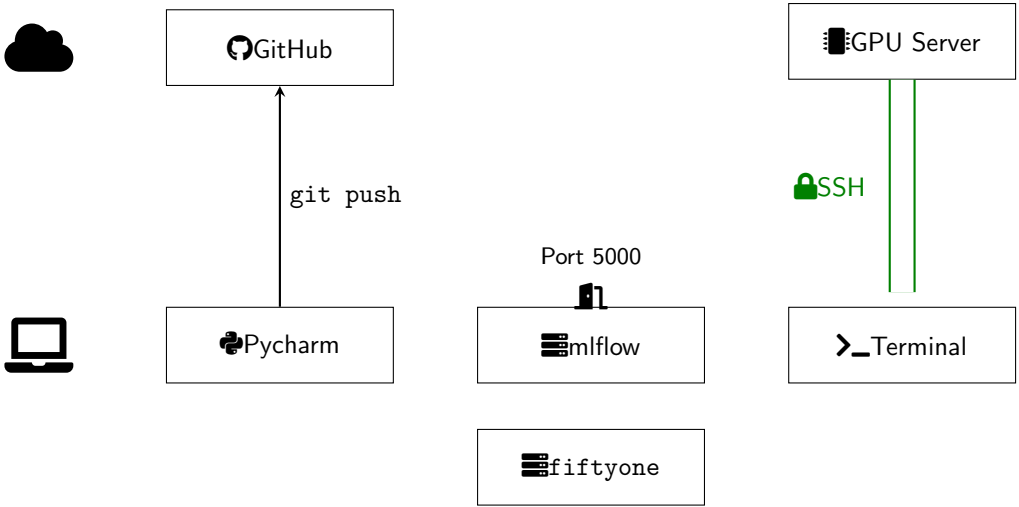


Figure: Example of data augmentation techniques applied to a CIFAR-10 image

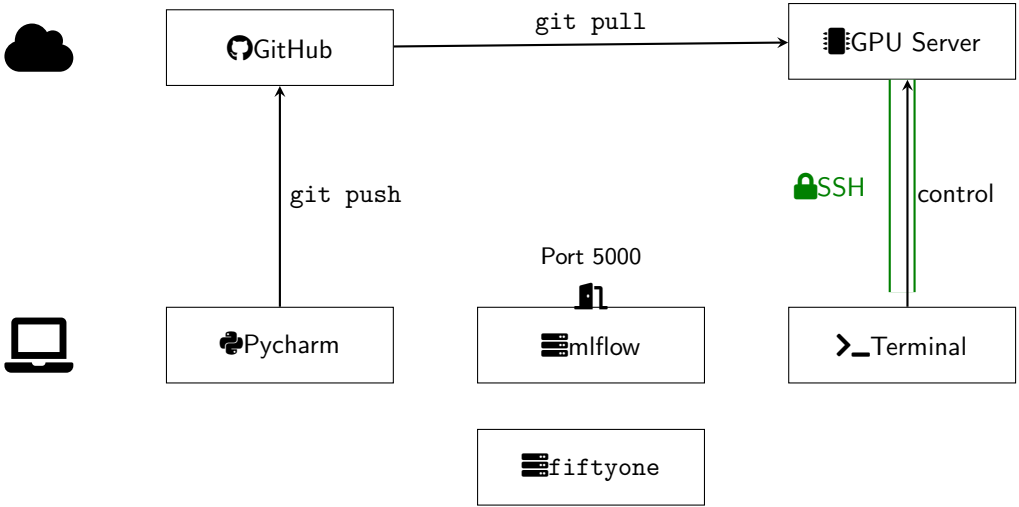
Training setup



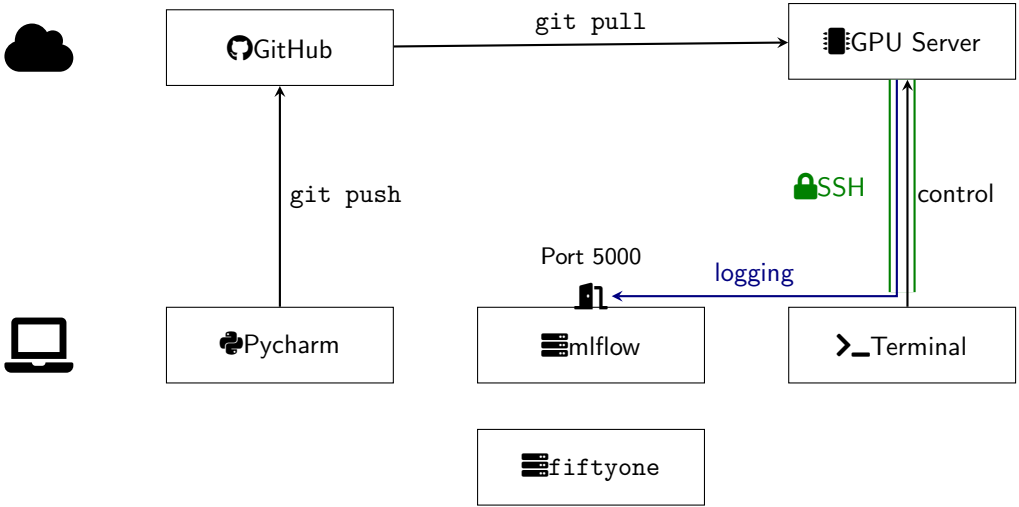
Training setup



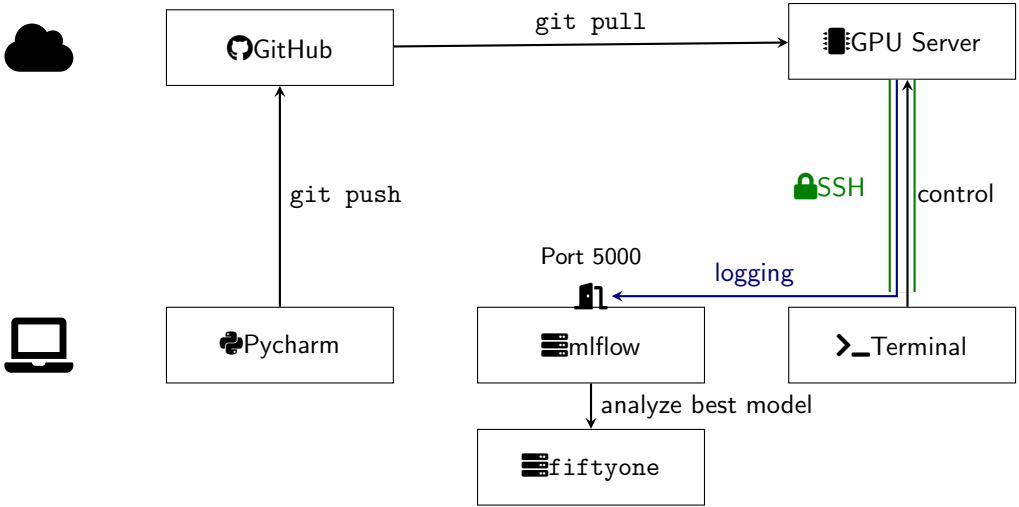
Training setup



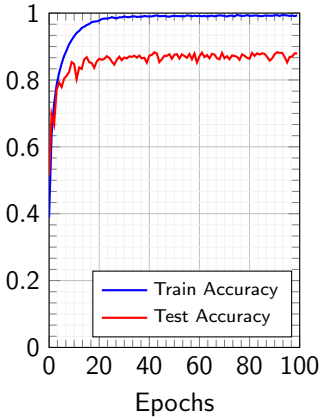
Training setup



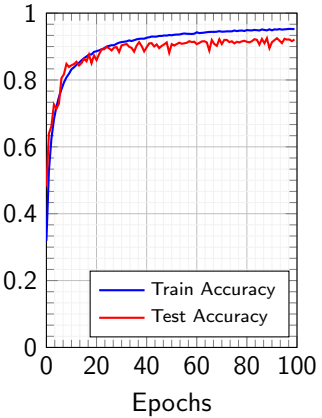
Training setup



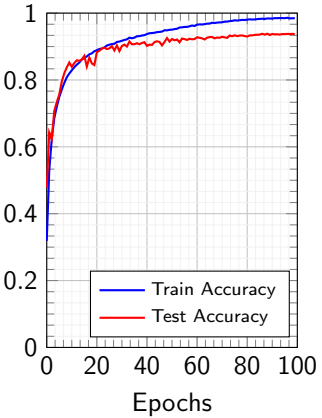
Results



(a) Accuracy in vanilla training



(b) Accuracy with data augmentation



(c) Accuracy with lr scheduling

Figure: Training performance

References I

Dosovitskiy, Alexey et al. (2021). *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. arXiv: 2010.11929 [cs.CV]. URL: https://arxiv.org/abs/2010.11929.

Krizhevsky, Alex (2009). "Learning Multiple Layers of Features from Tiny Images". In: URL: https://api.semanticscholar.org/CorpusID:18268744.